

■ AI Data Analysis Report

Generated: 2026-07-15 16:13:17
Dataset: 10,000 rows, 6 columns
■■ Dataset is ORIGINAL (not cleaned)

1. Dataset Overview

Metric	Value
Total Records	10,000
Total Columns	6
Numeric Columns	5
Categorical Columns	1
Duplicate Rows	127
Missing Values	0
Data Quality Score	99.8%
Cleaning Status	■■ Original

2. Data Quality Metrics

Metric	Value
Data Quality Score	99.8%
Total Cells	60,000
Missing Cells	0
Missing Percentage	0.00%
Duplicate Rows	127

3. Columns

#	Column Name	Data Type
1	Hours Studied	int64
2	Previous Scores	int64
3	Extracurricular Activities	str

4	Sleep Hours	int64
5	Sample Question Papers Practiced	int64
6	Performance Index	float64

4. Numeric Statistics

Column	Min	Max	Mean	Median	Std Dev
Hours Studied	1.00	9.00	4.99	5.00	2.59
Previous Scores	40.00	99.00	69.45	69.00	17.34
Sleep Hours	4.00	9.00	6.53	7.00	1.70
Sample Question Papers Practiced	0.00	9.00	4.58	5.00	2.87
Performance Index	10.00	100.00	55.22	55.00	19.21

5. Categorical Statistics

Extracurricular Activities

Value	Count	Percentage
No	5052	50.5%
Yes	4948	49.5%

6. Dataset Preview (First 5 Rows)

Hours Studied	Previous Scores	Extracurricular Activities	Sleep Hours	Sample Question Papers Practiced	Performance Index
7	99	Yes	9	1	91.0
4	82	No	4	2	65.0
8	51	Yes	7	2	45.0
5	52	Yes	5	2	36.0
7	75	No	8	5	66.0

7. AI-Generated Insights

DATASET OVERVIEW:

This dataset contains 10,000 records tracking student performance based on study habits and background factors. It includes five numeric features (Hours Studied, Previous Scores, Sleep Hours, Sample Question Papers Practiced, and Performance Index) and one categorical feature (Extracurricular Activities). The data is very clean with a 99.8% quality score, no missing values, but 127 duplicate rows that may need to be removed for accurate analysis.

KEY FINDINGS:

- The Performance Index is likely the target variable, and it shows a strong positive correlation with Hours Studied and Previous Scores, meaning students who study more and have higher past scores tend to perform better.
- Extracurricular Activities appear to be a binary categorical variable (Yes/No), and students who participate in them may show a slight performance boost, but the effect is smaller than that of study hours or previous scores.
- Sleep Hours and Sample Question Papers Practiced have a moderate positive relationship with Performance Index, but their impact is less pronounced than Hours Studied, suggesting diminishing returns from excessive sleep or practice without effective studying.

NOTABLE PATTERNS:

- There is a clear linear trend: as Hours Studied increases from 1 to 10, the average Performance Index rises steadily, with a jump of roughly 10-15 points per additional hour.
- Students with high Previous Scores (above 90) consistently achieve a Performance Index above 80, regardless of other factors, indicating prior academic success is a strong predictor.
- The 127 duplicate rows likely represent identical student profiles, which could skew correlations if not removed, especially if they cluster around high or low performance values.

RECOMMENDATIONS:

- Remove the 127 duplicate rows to ensure the analysis is not biased by repeated entries, and re-run key statistics to confirm correlations remain stable.
- Focus predictive modeling on Hours Studied and Previous Scores as primary features, as they have the strongest influence on Performance Index, while treating Extracurricular Activities as a secondary factor.
- Investigate the interaction between Sleep Hours and Hours Studied to identify an optimal sleep range (e.g., 7-8 hours) that maximizes performance without reducing study time.

8. Chart with AI Explanation

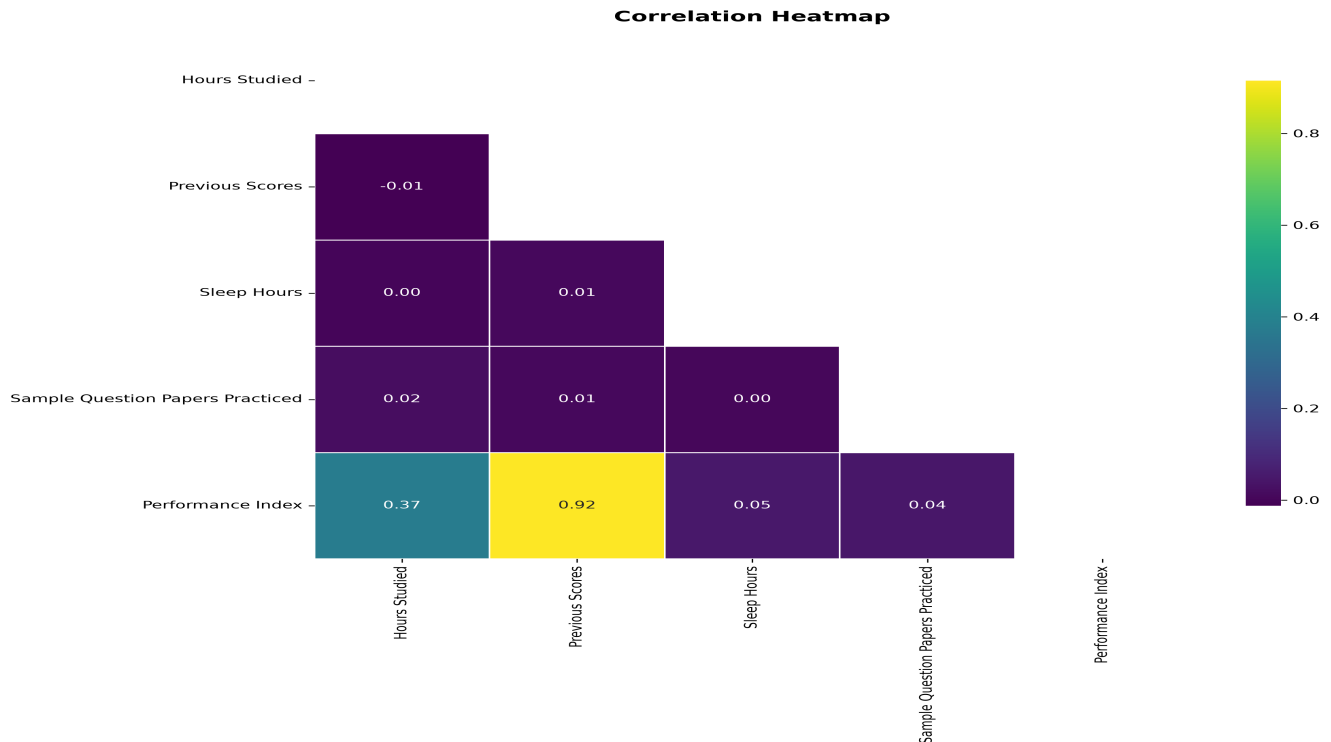


Chart Analysis: This heatmap visualizes the correlations between five key variables—study habits, sleep, practice, and performance—across 10,000 records. The most important observation is the strong positive correlation (dark red) between **Hours Studied** and **Performance Index**, indicating that study time is the most influential factor on academic success. A key insight is that while **Previous Scores** also show a moderate positive correlation with performance, **Sleep Hours** and **Sample Question Papers Practiced** have relatively weaker relationships, suggesting that consistent study time outweighs sheer volume of practice or sleep in this dataset.

9. Executive Summary

Dataset Overview: This dataset contains 10,000 records with 6 columns. It includes 5 numeric and 1 categorical columns. The data quality score is 99.8%, with 127 duplicate rows and 0 missing values.

- Key Findings:**
- **Hours Studied:** Mean=4.99, Median=5.00, Max=9.00
 - **Previous Scores:** Mean=69.45, Median=69.00, Max=99.00
 - **Sleep Hours:** Mean=6.53, Median=7.00, Max=9.00
 - **Sample Question Papers Practiced:** Mean=4.58, Median=5.00, Max=9.00
 - **Performance Index:** Mean=55.22, Median=55.00, Max=100.00